CMU 10-703: Deep Reinforcement Learning

Lecture 13: Offline Reinforcement Learning

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Offline Reinforcement Learning

Input
A fixed dataset of transitions collected before training.

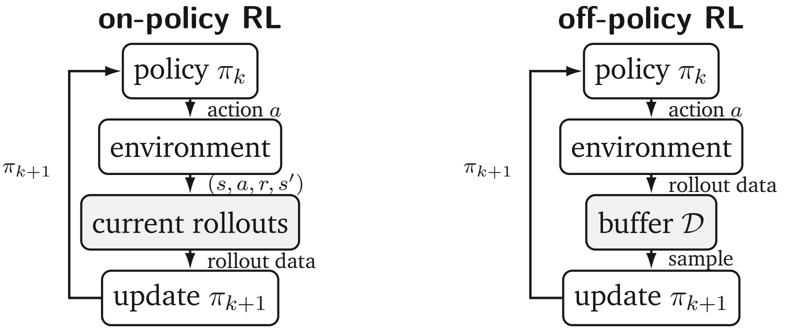
Goal
Learn a policy that improves on the data-collection policy.

Constraint
No environment interaction can correct training errors.

$$\mathcal{D} = \{(s_i, a_i, r_i, s_i')\}_{i=1}^{N}$$

Offline RL must improve from recorded experience without treating unsupported actions as reliable.

Online and Off-Policy Interaction



On-policy

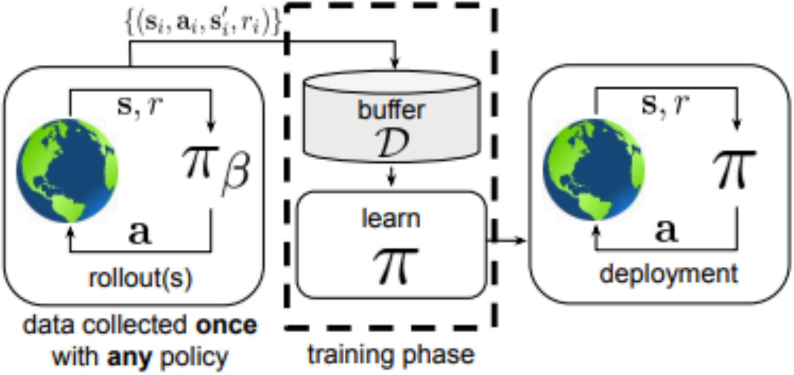
Each policy update uses rollouts from the current policy.

Online off-policy

A replay buffer stores past data, but interaction continues.

Source: Levine et al., NeurIPS 2020 Offline Reinforcement Learning tutorial.

Offline Training Pipeline



Data collection ends before learning begins; the trained policy interacts with the environment only at deployment.

Source: Levine et al., NeurIPS 2020 Offline Reinforcement Learning tutorial.

Three Data Regimes

Online, on-policy

Current policy collects current data.

$\pi_{\text{data}} = \pi_{\text{learned}}$

Online, off-policy

A behavior policy fills a replay buffer while training continues.

$\pi_{\text{data}} \neq \pi_{\text{target}}$

Offline

A fixed dataset is the only training input.

$\mathcal{D}$ never changes

Offline RL is off-policy learning with data collection completely finished.

1.

Offline RL Setup

Fixed Transition Dataset

Offline dataset. A static collection of recorded transitions, possibly augmented with termination flags or other logged fields.

$$\mathcal{D} = \{(s_i, a_i, r_i, s_i')\}_{i=1}^{N}$$

Dynamics
Successor states reveal what followed recorded actions.

Rewards
Logged rewards evaluate recorded outcomes.

Counterfactual gap
The data do not reveal every unchosen action's outcome.

The dataset is the only evidence about the environment during training.

Behavior Policy

Behavior policy. The policy π_β generated the actions in the fixed dataset.

behavior policy π_β $\rightarrow$ environment $\rightarrow$ fixed dataset $\mathcal{D}$ $\rightarrow$ learned policy π

Often unknown
The learner may observe only the consequences of π_β's decisions.

Not necessarily expert
The data may mix effective, poor, and random behavior.

Behavior Policy Sources

Human operation
Teleoperation logs from surgery, grasping, driving, or other tasks.

Previous agents
Experience recorded from an older or suboptimal policy.

Exploration
Random or deliberately diverse actions collected for coverage.

$$(s, a) \sim d^{\pi_\beta}(s)\pi_\beta(a \mid s)$$

The behavior policy determines which state–action pairs the learner can evaluate from evidence.

Dataset Coverage

Coverage. The dataset has coverage where it contains enough evidence to estimate outcomes for relevant state–action pairs.

well-covered pairs → **reliable estimates** → supported improvement

Finite data
Many possible state–action pairs may never appear.

Policy mismatch
A learned policy can select actions unlike those in $\mathcal{D}$.

Limited coverage sets the central statistical constraint of offline RL.

2.

Motivation and Objective

Motivations for Offline Learning

Safety-critical interaction
Public-road driving and physical robots cannot explore through costly failures.

Expensive experiments
Drug discovery and materials science may require slow synthesis, simulation, and testing.

Existing data
Passive logs can be repurposed, much as internet-scale text became training data for language models.

Offline training avoids repeated interaction when trial and error is unsafe, slow, or impractical.

A learned offline policy can also provide a safer initialization for cautious online fine-tuning.

Data-Driven RL Objective

Offline RL objective. Use the fixed dataset to learn a policy that maximizes expected discounted return in the underlying MDP.

$$\max_{\pi} \mathbb{E}_{s_t, a_t \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

Optimization target

Return under the learned policy π.

Available evidence

Transitions generated by π_β.

The objective evaluates π, while the dataset was sampled under π_β.

Offline RL and Imitation Learning

Imitation learning

$$\min_{\pi} \; D(\pi(\cdot \mid s) \,\|\, \pi_\beta(\cdot \mid s))$$

Reproduce the behavior distribution, usually assuming expert or near-expert data.

Offline reinforcement learning

$$\max_{\pi} \; J(\pi)$$

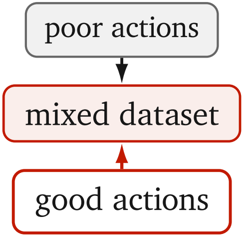
Use rewards and dynamics to improve on behavior that may be mixed or suboptimal.

Offline RL seeks supported improvement, not faithful reproduction of π_β.

3.

Sources of Improvement

Learning from Good Actions



Selection through value
Reward information can distinguish useful actions from poor ones within the same dataset.

With sufficient coverage of all state–action pairs, even uniformly random data can support recovery of an optimal policy by value-based methods.

Improvement is possible when the dataset contains good decisions that the learner can identify and reuse.

Generalization Beyond Recorded States

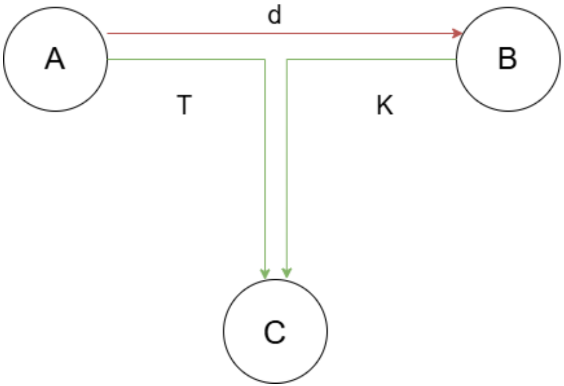
recorded state–action pairs → **function approximation** → similar unseen pairs

Local transfer
Deep models can share statistical strength across similar inputs.

Coverage remains necessary
Predictions far from the data manifold are not justified by nearby examples.

Generalization can extend coverage locally, but it does not make arbitrary counterfactual actions reliable.

Trajectory Stitching



Recorded segments

The dataset contains trajectories from A to C and from B to C.

Learned route

Value-based reasoning can combine useful segments into a direct policy from A to B.

Imitation alone reproduces recorded trajectories rather than optimizing their composition.

Source: Levine et al., NeurIPS 2020 Offline Reinforcement Learning tutorial.

4.

Distribution Shift

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Counterfactual Actions

Offline RL must evaluate actions that may be weakly represented or absent in the dataset.

$$\text{actions in } \mathcal{D} \xrightarrow{\text{fit}} \textbf{optimize } \pi \xrightarrow{\text{may select}} \text{unsupported actions}$$

Near the data manifold
Generalization may be accurate enough for comparison.

Far from the data manifold
Values and rewards can be arbitrary extrapolations.

Distribution Shift

Distribution shift. Training data follow the behavior distribution, while policy optimization queries a changing learned distribution.

$$(s, a) \sim d^{\pi_\beta}(s)\pi_\beta(a \mid s) \qquad \text{versus} \qquad (s, a) \sim d^{\pi}(s)\pi(a \mid s)$$

Policy optimization
Samples actions from a learned π that changes during training.

Value maximization
Queries $\max_a Q(s, a)$, which searches for high predictions.

Optimization can move toward actions where prediction error is largest.

Bandit Warm-Up

Fit a reward model on logged actions

$$\min_{\theta}\ \mathbb{E}_{(s,a,r)\sim\mathcal{D}}\left[\left(r_\theta(s,a)-r\right)^2\right]$$

In-domain predictions are trained against observed rewards.

Bandit Warm-Up

Fit a reward model on logged actions

$$\min_{\theta} \; \mathbb{E}_{(s,a,r)\sim\mathcal{D}} \left[\Big(r_\theta(s,a) - r \Big)^2 \right]$$

Optimize a policy against the learned model

$$\max_{\pi} \; \mathbb{E}_{\substack{s\sim\mathcal{D} \\ a\sim\pi(\cdot|s)}} \left[r_\theta(s,a) \right]$$

The policy searches beyond the actions used to fit the model.

Bandit Warm-Up

Fit a reward model on logged actions

$$\min_{\theta} \mathbb{E}_{(s,a,r)\sim\mathcal{D}}\left[\left(r_\theta(s,a) - r\right)^2\right]$$

Optimize a policy against the learned model

$$\max_{\pi} \mathbb{E}_{\substack{s\sim\mathcal{D} \\ a\sim\pi(\cdot|s)}}\left[r_\theta(s,a)\right]$$

Positive out-of-distribution errors become attractive actions, even when their true rewards are poor.

Out-of-Distribution TD Targets

Temporal-difference loss on recorded transitions

$$\mathcal{L}_{\mathrm{TD}}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(r + \gamma \max_{a'} Q_{\mathrm{tgt}}(s', a') - Q_\theta(s, a)\right)^2\right]$$

The fitted pair (s, a) is observed.

Out-of-Distribution TD Targets

Temporal-difference loss on recorded transitions

$$\mathcal{L}_{\mathrm{TD}}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(r + \gamma \max_{a'} Q_{\mathrm{tgt}}(s',a') - Q_\theta(s,a)\right)^2\right]$$

The loss uses in-dataset (s, a), but its target maximizes over actions a' that need not be in the dataset.

The backup can query an unsupported next action a'.

Out-of-Distribution TD Targets

Temporal-difference loss on recorded transitions

$$\mathcal{L}_{\text{TD}}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(r + \gamma \max_{a'} Q_{\text{tgt}}(s', a') - Q_\theta(s,a)\right)^2\right]$$

The loss uses in-dataset (s, a), but its target maximizes over actions a' that need not be in the dataset.

A spurious positive peak enters the target for an otherwise supported transition.

OOD Error Amplification by Bootstrapping

$\boxed{\text{OOD action } a'}$

Maximization searches among unsupported next actions.

OOD Error Amplification by Bootstrapping

OOD action a' → positive Q-error

A positive approximation error wins the maximization.

OOD Error Amplification by Bootstrapping

OOD action a' → positive Q-error → **inflated TD target**

The erroneous value enters a target for a recorded transition.

OOD Error Amplification by Bootstrapping

OOD action a' → positive Q-error → **inflated TD target** → inflated data values

extrapolation error $\longrightarrow$ backup error $\longrightarrow$ value blow-up

Repeated backups propagate the error through the dataset.

Online Correction and Offline Persistence

Online off-policy RL

chase high estimate → collect data there → correct the estimate

Offline RL

chase high estimate → no new transition → error persists

A fixed dataset cannot realign the behavior and learned distributions after the policy drifts.

Failure Symptoms

Critic failure

- Q-value overestimation
- Growth or divergence with more gradient steps
- Corrupted targets for covered actions

Policy failure

- Collapse to narrow OOD modes
- High predicted return
- Poor return in the true MDP

Double critics and ensembles can slow overestimation, but they do not remove the unsupported OOD query.

PPO Importance Ratios

PPO update. Compare the learned policy with the policy μ associated with the sampled action, then clip the ratio.

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\mu(a_t \mid s_t)}, \qquad r_t(\theta) \in [1-\epsilon, 1+\epsilon]$$

Clipping controls

The size of a policy update relative to μ.

Clipping does not control

Whether the update points toward supported actions.

Limitations of Plain PPO

$\mu = \pi_\beta$

- Improvement requires departure from behavior.
- The ratio reaches the clip range quickly.
- Gradients vanish and learning stalls.

$\mu =$ **stale** π_θ

- Successive steps can drift from the dataset.
- OOD value errors still direct the update.
- Small unsupported steps accumulate.

Plain PPO either under-updates against fixed behavior or drifts through unsupported policy iterations.

5.

Offline RL Remedies

Policy Constraints

Support matching. Regularize the learned policy toward the behavior distribution while still optimizing value.

$$\max_{\pi} \mathbb{E}_{\substack{s\sim\mathcal{D} \\ a\sim\pi(\cdot|s)}} [Q_\theta(s,a)] - \alpha D(\pi(\cdot \mid s) \,\|\, \pi_\beta(\cdot \mid s))$$

Possible constraints

KL divergence, behavior cloning loss, or explicit support restrictions.

Effect

The actor proposes fewer unsupported actions for optimization and targets.

Pessimistic Value Learning

Conservative critic. Penalize values on actions sampled from a distribution with broader support than the behavior data.

$$\min_{\theta} \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(Q_\theta - (r + \gamma \widetilde{V})\right)^2\right]}_{\text{TD fit on data}} + \lambda \underbrace{\mathbb{E}_{\substack{s\sim\mathcal{D} \\ a\sim\nu(\cdot|s)}}[Q_\theta(s,a)]}_{\text{conservative penalty}}$$

Broader action distribution
ν may be uniform, the actor, or a mixture.

Effect
Unsupported actions receive lower values and produce safer targets.

Two Remedy Families

Constrain the policy

Keep π near the action distribution represented in $\mathcal{D}$.

$$\pi \approx \pi_\beta$$

Constrain the values

Make unsupported actions unattractive to the policy.

$Q(s, a_{\mathrm{OOD}})$ is pessimistic

The two remedies are complementary and can be used with value-based, actor–critic, policy-gradient, or model-based methods.

Offline RL Workflow

collect diverse data → train conservatively → evaluate offline → deploy cautiously

Coverage
Include diverse states and actions when data collection is possible.

Validation
Use off-policy evaluation and conservative model selection.

Fine-tuning
If interaction is allowed, begin cautious online learning from the offline policy.

Offline Reinforcement Learning Summary

Problem
Learn from a fixed dataset collected by an unknown or suboptimal behavior policy.

Opportunity
Select good actions, generalize locally, and stitch useful trajectory segments.

Failure mode
Distribution shift lets policy optimization exploit OOD prediction errors.

$$\begin{aligned}\text{reliable offline improvement} &= \text{supported policy search} \\ &\quad + \text{conservative value estimation}\end{aligned}$$

Offline RL improves on logged behavior only to the extent that its policy and value estimates respect the evidence in the dataset.